

1. Introduction

1.1 Purpose

This SRS describes a weather forecasting system that can assist farmers and agriculture companies.

This is the first release of a document; no revision has been done so far.

This SRS describes a full system that uses external solutions, such as IBM's GRAF model, Davis Vantage Pro-2 weather station, and many more.

1.2 Document Conventions

This document follows the SRS standard structure provided by the IEEE computing society.

Titles and subtitles are highlighted in bold. Text is written in the normal form, but details that require special attention are highlighted in bold, these details can be standards, definitions or important points that address a sub-section objective (or vital to the whole SRS.)

We suggest you start with the second section, in particular sub-sections **2.1, 2.2, 2.4**, (if you are a developer, sub-section 2.5 is relevant.).

Then to go through Functional requirements (in case of implementation) and Non-functional requirements (**To see if we can proceed with the project. For example, if the standards we need to adhere to are hard to satisfy financially, then the whole project is unattainable.**).

Finally you can end up with the third section.

Reviewing the UML diagrams can be done with all the sections to help understand their aim.

1.3 Intended Audience and Reading Suggestions

Since the document is prepared by students interested in the software process, the document is helpful for other students as a reference to help them understand the process of requirement elicitation and analysis.

The document can also be used by developers who wish to realize the specified solution in the document, although we do not expect a developer to be able to implement it alone but a complete enterprise.

1.4 Product Scope

The Weather Prediction System here is providing an application aimed at helping farmers get specific location weather reports using IoT devices, AI models, and weather stations. Because it provides predictions on the weather conditions, it assists farmers in making critical decisions.

This system is designed to raise the efficiency of farming, restrain the amounts of losses, increase the yield, and as a result increase the economic benefit coming out of the farming process.

1.5 References

- **“5G Wireless: A Comprehensive Introduction 1st Edition, By Dr. William Stallings”, this book helped us with all of the external solutions that are proposed to be used in the system.
5G, edge nodes, data centers, IoT devices are all ideas derived from the book, we owe a lot of the SRS content to it.**
-
- IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications.
- Davis Vantage Pro 2 Weather Station, “[Vantage Pro2 — Davis Instruments](#)”.
- WeatherLink cloud solution, Davis Instruments, “[WeatherLink Home](#)”.
- IBM AI model, GRAF. “[GRAF Weather Model | High Resolution Weather Data Forecast](#)”.

2. Overall Description

2.1 Product Perspective

The solution described in this document is new in the way that it integrates different components, these components can be used as standalone systems in weather forecasting or other application areas, of course, these solutions are provided by third-party vendors. The idea here is to combine all of these “sub-systems” to achieve the goal of giving an accurate forecast.

As stated earlier, we rely heavily on solutions provided by different vendors, whether it is the Pro-2 weather station which is supplied by Davis corporate, or GRAF AI model from IBM, **Thus the solution is not self-contained.**

Our work on the project is about the successful integration of these components and building the app that a farmer is going to use.

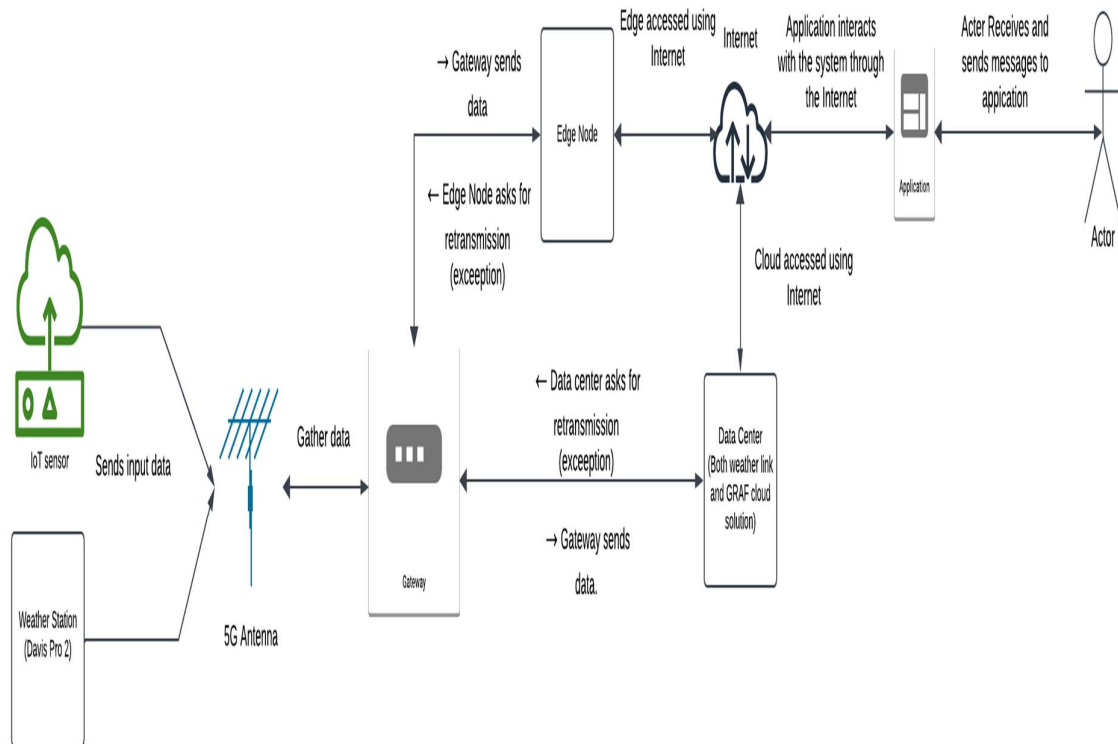


Figure: Context Diagram

2.2 Product Functions

The following list outlines the key functions and features of the system, organized by user roles and responsibilities:

- Create an Account
- Manage Account
- Delete Account
- Login/Logout
- Ask for Help
- Search for Weather Forecast
- View Weather Details
- Customize Forecast Parameters
- Set Weather Alert Thresholds
- Receive Notifications for Alerts
- Generate Historical Weather Reports
- Download Reports in Various Formats
- Integrate sensor data (for developer)
- View Sensor Data (for administrator)
- Manage IoT Sensor Integration (for developer)
- Place Weather Data Queries (done by the system upon farmer request)
- Visualize Data with Graphs and Tables (done by the system upon farmer request)

2.3 Operating Environment

The system requires the following hardware components:

IoT device: Many sensor types differ in price and capabilities. The decision on which device to use will be determined according to the feasibility analysis.

Weather station: Davis Vantage Pro 2, works along with sensors to diversify our source of input data.

Farm Gateway: Our hardware device is responsible for collecting input data and transmitting it to the data center (WeatherLink by Davis) or the edge node.

The system must be supported with a 5G infrastructure; 5G will help us in promoting our solution since most farms are in rural areas, and the cost of establishing a wired network is too huge. In 5G, all we need to have been antennas.

Software components:

Data Center: WeatherLink by Davis, here we will process and store the data; We cannot run our AI model here since the cloud solution is not designed to handle such cases.

AI Model: GRAF from IBM is used to perform pattern recognition on the data, the model will run on a dedicated cloud solution.

2.4 Design and Implementation Constraints

The developer must realize that the sensors used in the system are too primitive to do any complex operation. So, the developer should not expect a sensor to retransmit a data packet due to some error.

In the system we are restricted with the API provided by Davis to interact with their cloud solution; this is also applicable to our AI model (GRAF) that runs on another data center.

The system must support parallel data transmission from its sensors.

We will have a lot of different data format standards between the different components we are using; this can be an obstacle.

GRAF model requires a platform that can support its computation.

As an entity (referring to us) providing the solution, we must comply with any constraints placed from the different external solution vendors.

2.5 User Documentation

We will create a video tutorial that shows the farmer how to use our system.

We will give quick training for administrators (since they work with the farmer), covering the necessary functionalities of the system.

We will also provide an administration document to an administrator (not to be confused with a technical document, it is confidential).

A bot assistant in the system application, to help the farmer with any questions.

2.6 Assumptions and Dependencies

A bullet point here describes both an assumption and a dependency, we concluded that the two points are referring to the other, so we combined them together.

- Pro 2 weather station and sensors are achieving their objective, which is to collect input data.
- A 5G service provider cannot establish the network in a rural area (typical for a farm) or provide the service but with high expenses.
- The edge node used in the system is helpful, the main objective of using the edge is to process repetitive tasks and make the data center do new complex tasks instead. The edge can be not helpful if the tasks are inherently complex.
- The assumption that we can maintain all these external solutions together brings to the table the importance of finding alternative solutions.

3. Other Nonfunctional Requirements

3.1 Performance Requirements

3.2 Safety Requirements

IoT devices and weather stations **shall** comply with standards ensuring their durability and ability to withstand harsh weather conditions.

The standard used here is IEC 61010-1.

IoT devices in the system **should** be designed to consume minimal energy to reduce the carbon footprint. **Our sensors will adhere to the Energy Star standard**, which outlines how to achieve low energy consumption in IoT devices.

Since our system will deploy 5G infrastructure, we **shall** establish a periodic monitoring system for electromagnetic radiation levels, ensuring that the radiation from the system does not exceed safe limits. We will base these limits on the recommendations of **ANSI** and **IEEE**.

Another potential hazard is fire, which could arise from any of the devices in the system. If one device were to catch fire, it could put the entire farm at risk. Our system **shall** adhere to the main standard for preventing such incidents, **UL 61010-1**.

3.3 Security Requirements

Data transmitted within the system shall be encrypted to prevent eavesdropping by adversaries. The encryption protocols used will be different depending on the communication context.

- For communication between the cloud and the farm gateway, we will implement **TLS (Transport Layer Security)**.
- For communication between sensors and gateways, we will implement **AES (Advanced Encryption Standard)** with a 256-bit key for end-to-end encryption.

The system shall require user authentication for access to sensitive data or control over devices.

The system shall implement **multi-factor authentication (MFA)**, to ensure a strong authentication process, a message to be sent to the user's phone asking for their confirmation.

The system shall support **Role-based access control (RBAC)**, this is obvious since we will have different users with different roles. A farmer must not have the privileges of an administrator who will be able to control IoT devices.

3.4 Software Quality Attributes

Scalability: the system shall be scalable, by increasing or decreasing the number of IoT devices or Pro 2 weather stations. **This attribute is quantifiable.**

Reliability: The system network shall be reliable; we expect a successful transmission of packets almost all the time (See non-functional requirements). **This attribute is quantifiable.**

Interoperability: the developer shall ensure that the system developed is interoperable. **This attribute is verifiable.**

Durability: The system shall withstand extreme weather conditions. **This attribute is verifiable.**

Maintainability: the developer shall make the system maintainable so that new enhancements can be integrated easily. **This attribute verifiable.**

Usability: the developer shall guarantee the satisfaction of the user from the system. Gathering feedback from users and having an intuitive user application design satisfies this attribute. **This attribute is verifiable.**

Robustness: the system shall be robust, recovering from failures seamlessly, such as loss of connection between the gateway and IoT devices or the shutdown of a gateway must be addressable. **This attribute is verifiable.**

3.5 Business Rules

In a business, everyone will have a set of accepted roles that he can play. The system shall allow only authorized users to do a task that is assigned to them, e.g., only Network administrators can configure the 5G infostructure.

In a business, it is important to conduct periodic monitoring of energy usage. The main objective is to achieve a feasible business process.

No maintenance activities can be done during critical farming periods, this is important to avoid disruptions of the system operations.
Sharing weather data with external entities (e.g., government) collected from different farms is legal. Farmers shall be informed that their farm's weather data is not confidential, and that the solution provider has the right to use the collected data as they see fit.

4. State Transition Diagram

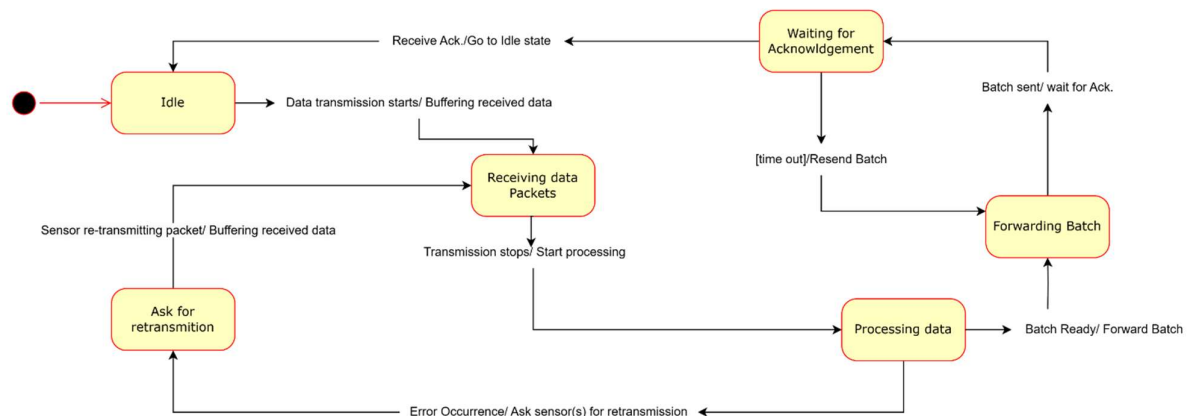


Figure: State Machine Diagram of a Gateway